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Mythbusting #5

## **“Computer Security Is Pretty Much Done”**

For decades we have been told that strong cryptography, secure operating systems, firewalls, and intrusion detection systems have “solved” computer security. The myth goes something like “We built the systems so the problem of cyberattacks is pretty much handled.” If anything is wrong now, it must be because someone forgot to turn something on. As Bruce Schneier puts it in *Secrets and Lies*, it's easy to believe this myth because “between system security measures... and good network security measures... It seems as if computer security is pretty much done.” If the theory is perfect, shouldn't the practice be close behind?

Imagine a brand new car with lane assist, automatic braking, and collision detection. The dealership advertises it as “safer than ever.” If a single sensor is misaligned, or a software update introduces a tiny bug, the whole system can behave unexpectedly. The safety design might be good, but the implementation can still fail.

Schneier uses the Ariane 5 disaster to show how even organizations with huge budgets and strong incentives for correctness still ship flawed code. A tiny data conversion bug destroyed a multibillion dollar rocket. The backup system failed the same way because it ran the same flawed software. This wasn't a security failure, but it illustrates the core point, that complex systems fail in unexpected ways, even when security is at its best. Attackers exploit exactly these kinds of subtle mistakes.

We see this myth in marketing all the time. “Military grade encryption,” “unhackable password manager,” “AI powered threat prevention.” Yet week after week, companies patch critical vulnerabilities in these same products. The design may be strong, but the implementation is full of cracks. Class lectures on human factors emphasized that people tend to overestimate the power of tools and underestimate the messy reality of implementation. If a system looks sophisticated, people assume it is secure. Schneier calls this the gap between “theory” and “practice.” Cryptography, secure kernels, and firewalls work well on paper, but real systems are built out of imperfect code, rushed development cycles, and constant feature updates. The myth comes from misplaced trust in the elegance of the design, not the reality of the implementation.

Believing that “computer security is done” leads to harmful decisions at every level:

Organizations underinvest in security engineering

- If leaders believe this myth, they don't fund code audits, secure development training, or long testing cycles. They assume the tools they bought are enough. This is how vulnerabilities go unnoticed for years.

Developers rush features instead of fixing bugs

- Schneier notes that companies often release software before fixing known bugs because of market pressure. If security is “done,” why delay a release for something that probably won’t fail? This mindset leads to large failures, such as the Cisco switch bug that knocked out AT&T’s network for 6,600 customers.

Users develop false confidence

- People assume that because their device has a firewall and encryption, they can click any link or reuse any password. When the myth tells them the system is secure, they stop behaving securely.

Policymakers misunderstand the problem

- If security is “done,” then breaches must be due to individual errors. This leads to user blaming or assuming that buying a new product will magically fix systemic issues.

Security is not even close to being “done.” Schneier argues that implementation flaws are far more common and far more dangerous than design flaws. Real systems are too complex to test extensively, built under time pressure, constantly changing, and attacked by intelligent attackers who deliberately seek out rare edge cases. A more accurate belief is that computer security is an ongoing process, not a finished product. It is not something that you install once, it’s something that you maintain, test, and update. We should assume that systems have bugs, that attackers will continue to advance with systems, and be skeptical of “unhackable” claims.